King Fahd University of Petroleum and Minerals
Department of Mathematics
Student effort not Official. Generated by Ice ❄. دعواتكم له

Math 201 Major Exam II Term 231

Original exam date: 14 November 2023. Practice version: questions and choices shuffled.
Net time allowed: 120 minutes

---

1. $\lim_{(x,y)\to(0,0)} \frac{e^{-x^2-y^2}-1}{x^2+y^2} =$

(a) does not exist

(b) -2

(c) 0

(d) -1

(e) 3

2. The range of

$$f(x,y) = 4\ln(3-2x^2-y^2)$$

is

(a) $(-\infty,\infty)$

(b) $[-1,\ln 4)$

(c) $(-1,1)$

(d) $(-\ln 4, 0]$

(e) $(-\infty, 4\ln 3]$

3. The level surfaces of $f(x,y,z) = \ln(4x^2+4y^2-z^2)$ are

(a) hyperbolic paraboloids

(b) hyperboloids of one sheet

(c) hyperboloids of two sheets

(d) elliptic cones

(e) ellipsoids

4. If the line with symmetric equations

$$x - 2 = z - 1, \qquad y = 1$$

is perpendicular to the surface $3x^2 - 2xy + z^2 = 1$ at the point (x_0, y_0, z_0), then $x_0 + y_0 + z_0 =$

(a) 2

(b) 1

(c) 0

(d) 4

(e) 3

5. The area of the region that lies in the first quadrant as well as in the domain of the function $f(x, y) = \ln(6 - x - y)$ is equal to

(a) 34

(b) 18

(c) 30

(d) 23

(e) 12

6. Let

$$f(x,y) = \begin{cases} \dfrac{\sin(xy)}{xy}, & xy \neq 0 \\ c, & xy = 0 \end{cases}$$

The value of c that makes $f(x, y)$ continuous at $(0, 0)$ is

(a) -4

(b) 1

(c) 2

(d) -1

(e) 0

7. Consider the surface

$$x^2 - 3y^2 - 9z^2 = 0.$$

Which of the following is/are correct? (I) The vertical trace in the xz-plane is the lines $x = \pm 3z$. (II) The traces in the plane parallel to the yz-plane are ellipses. (III) The surface represents a hyperboloid of two sheets.

(a) (I) and (II) only

(b) (II) only

(c) (I) only

(d) (III) only

(e) (II) and (III) only

8. If the directional derivative of $f(x, y) = \ln(1 + xy)$ at $(a, 4)$ in the direction of $\vec{v} = \langle 1, -2 \rangle$ is 0, then $a =$

(a) 2

(b) 0

(c) 4

(d) 1

(e) -1

9. If the radius and height of a right circular cone are measured as 10 cm and 25 cm, respectively, with a possible error in measurement of as much as 0.1 cm in each. The maximum error in the calculated volume of the cone is estimated as (Hint: The volume V of a cone with radius r and height h is $V = \frac{1}{3}\pi r^2 h$).

(a) 12π

(b) 4π

(c) 20π

(d) 10π

(e) 2π

10. Let $z = \tan^{-1}(xy)$ where $x = s + \sin t$ and $y = \cos t$. Then $\dfrac{\partial z}{\partial s} + \dfrac{\partial z}{\partial t}$ at $(s, t) = (1, 0)$ is equal to

(a) 0

(b) 8

(c) 1

(d) -4

(e) -2

11. Let $f(x, y, z) = xe^{xy} \sin^2 z$. Then $f_{xyz}(-1, 0, \pi/4) =$

(a) 0

(b) -1

(c) -2

(d) 4

(e) 1

12. The distance between the parallel planes

$$10x + 2y - 2z = 5 \qquad \text{and} \qquad 5x + y - z = 1$$

is equal to

(a) $\dfrac{1}{\sqrt{2}}$

(b) $\dfrac{\sqrt{3}}{6}$

(c) $\dfrac{1}{\sqrt{3}}$

(d) $\sqrt{13}$

(e) $\sqrt{7}$

13. $\lim_{(x,y)\to(0,0)} \frac{x^2 y}{x^4+y^2} =$

(a) does not exist

(b) 0

(c) -1

(d) 1

(e) ∞

14. If $2yz + x \ln y = z^2$, then $\dfrac{\partial z}{\partial y}$ at $(2, 1, 2)$ equals

(a) 3

(b) 1

(c) 5

(d) 4

(e) 0

15. If the equation of the plane containing the point $(1, 2, 3)$ and passing through the line of intersection of the planes

$$x + y + z = 1, \qquad 2x - y + 2z = 2$$

is given by $2x + ay + 2z = b$, then $a + b =$

(a) 3

(b) -2

(c) 0

(d) 2

(e) -1

16. Let $w = \sqrt{x^2 + y^2 + z^2}$, where

$$x = \cos\theta, \quad y = \sin\theta, \quad z = \tan\theta.$$

Then $\dfrac{dw}{d\theta}$ at $\theta = \dfrac{\pi}{4}$ is equal to

(a) $3\sqrt{2}$

(b) $\sqrt{3}$

(c) $\sqrt{2}$

(d) $2\sqrt{3}$

(e) 1

17. Let $W(s, t) = F(u(s, t), v(s, t))$, where F, u and v are differentiable functions, and

$$F_u(2, -3) = 1, \quad F_v(2, -3) = 3, \quad u(1, 0) = 2, \quad v(1, 0) = -3,$$

$$u_s(1, 0) = -2, \quad v_s(1, 0) = 5, \quad u_t(1, 0) = -6, \quad v_t(1, 0) = 4.$$

Then $W_s(1, 0) + W_t(1, 0) =$

(a) 0

(b) 22

(c) 11

(d) 20

(e) 19

18. If the line that passes through the points $P(1,3,4)$ and $Q(0,5,7)$ passes also through the point $R(2,1,a)$, then $a =$

(a) 1

(b) 5

(c) 3

(d) 2

(e) -2

19. If the maximum increase of the function $f(x,y) = x^2 + y^2 - 4x - 2y$ at the point $(5,c)$ occurs in the direction of $\vec{i} + 2\vec{j}$, then $c =$

(a) 7

(b) 4

(c) 0

(d) 5

(e) 3

20. The normal line to the surface

$$\ln\left(\frac{x}{y-z}\right) = x - 1$$

at the point $(1,4,3)$ passes through the point

(a) $(0,3,3)$

(b) $(1,2,3)$

(c) $(1,3,4)$

(d) $(2,3,3)$

(e) $(1,2,1)$

Answer Key

Student effort not Official. Generated by Ice ❄. دعواتكم له

Q	Ans	Brief explanation
1	**(d)**	With $u = x^2 + y^2$: $\lim_{u\to 0} \frac{e^{-u}-1}{u} = -1$.
2	**(e)**	$u = 3 - 2x^2 - y^2 \in (0, 3]$, so the range is $(-\infty, 4\ln 3]$.
3	**(b)**	$4x^2 + 4y^2 - z^2 = e^k > 0$: hyperboloids of one sheet.
4	**(c)**	$\nabla F = \langle 6x - 2y, -2x, 2z\rangle \parallel \langle 1, 0, 1\rangle \Rightarrow x_0 = 0,\ y_0 = -z_0$; sum $= 0$.
5	**(b)**	Domain $x + y < 6$ in the first quadrant is a triangle with legs 6: area 18.
6	**(b)**	$\lim \frac{\sin(xy)}{xy} = \lim_{u\to 0} \frac{\sin u}{u} = 1$, so $c = 1$.
7	**(a)**	Elliptic cone $x^2 = 3y^2 + 9z^2$: (I) true, (II) true, (III) false.
8	**(a)**	$\nabla f \cdot \langle 1, -2\rangle = \frac{4-2a}{1+4a} = 0 \Rightarrow a = 2$.
9	**(c)**	$dV = \frac{2}{3}\pi rh\,dr + \frac{1}{3}\pi r^2\,dh$ with $dr = dh = 0.1$ gives 20π.
10	**(c)**	$z_x = z_y = \frac{1}{2}$ at $(1, 1)$; $z_s = \frac{1}{2}$, $z_t = \frac{1}{2}$; sum $= 1$.
11	**(c)**	$f_{xy} = xe^{xy}(2 + xy)\sin^2 z$, $f_{xyz} = xe^{xy}(2 + xy)\sin 2z = -2$ at the point.
12	**(b)**	Same normal: $D = \frac{\lvert 5-2\rvert}{\sqrt{108}} = \frac{\sqrt{3}}{6}$.
13	**(a)**	Along $y = 0$ the limit is 0, along $y = x^2$ it is $\frac{1}{2}$: does not exist.
14	**(a)**	$F = 2yz + x\ln y - z^2$: $z_y = -\frac{F_y}{F_z} = -\frac{6}{-2} = 3$.
15	**(e)**	Pencil $P_1 + kP_2 = 0$ with $k = -\frac{5}{4}$ gives $2x - 3y + 2z = 2$: $a + b = -1$.
16	**(c)**	$w = \sqrt{1 + \tan^2\theta} = \sec\theta$, so $w' = \sec\theta\tan\theta = \sqrt{2}$ at $\pi/4$.
17	**(e)**	$W_s = 1(-2) + 3(5) = 13$, $W_t = 1(-6) + 3(4) = 6$, sum $= 19$.
18	**(a)**	Line $(1 - t,\ 3 + 2t,\ 4 + 3t)$; $x = 2 \Rightarrow t = -1$ ($y = 1$ checks), so $a = 1$.
19	**(a)**	$\nabla f(5, c) = \langle 6, 2c - 2\rangle \parallel \langle 1, 2\rangle \Rightarrow 2c - 2 = 12$, $c = 7$.
20	**(c)**	$\nabla F(1, 4, 3) = \langle 0, -1, 1\rangle$; line $(1, 4 - t, 3 + t)$ passes through $(1, 3, 4)$.